Rank-27 Kernel Import-Rank Representation Repair

Tactix Development Project

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Executive summary

Matched current-Tactix/current-Kissat Substitute localization assigned the first portable same-work cost gap to reconstructing permanent first-import order through the external-variable map on every hot comparison. The accepted repair preserves that permanent external rank as proof and reconstruction authority while maintaining an equivalent direct image over current compact Kernel identities. First import writes both views, variable compaction remaps the direct view, and invariants assert equality. The fallback for unrecorded public-API variables remains stable external order.

The exact commit-relative candidate diff has SHA-256 `681d17c986180b99b3c4cc6eb830dc32e88ea6a5b1ab2f1233872bca786ee1b6`; the clean Release binary has SHA-256 `c7c46bfa4bf8571dda7b270008b9e800c87cd23074b25b40ced3bbae43a057d4`. At the exact conflict-9450 common state, import-order helper cycles fall 32.53% and complete prefix cycles fall 1.55%, with identical rule/effect state and a byte-identical 64-row event trace. Same-allocation full Rank 27 preserves the complete logical state and improves wall time 458.89 to 450.21 seconds (−1.89%). The complete B28+C5 gate passes 66/66 at 1.008483x predecessor PAR-2 with no solve loss, invalidity, or certificate residue. A separate Rank-27 proof witness verifies the complete 1.514 GB proof and preserves both the accepted proof hash and proof-mode logical identity.

The full Release suite is 73/74: every functional, solver, proof/model, trace, boundary, reset, cleanup, and hygiene test passes; only the existing opt-in documentation-size policy debt fails. Claude Code's authoritative exact-source fuzz report is PASS, completing the final correctness gate.

1 Localization, rejected arm, and source boundary

Matched Substitute campaign `1647662` compares completed invocations from the same conflict-9450 state. Tactix spends 1766 ms in Substitute versus 931.761 ms in current instrumented Kissat. SCC discovery is not the owner: 601 ms versus 570.808 ms. Tactix's rewrite and maintenance tail costs 1159 ms versus about 361 ms; its measured substages include 221 ms snapshot, 265 ms binary rewrite, 190 ms large rewrite, 34 ms post-work, and 419 ms collection. Source and sampled-cycle tracing identify repeated `kernel_import_var_order` translation as the earliest bounded representation owner.

The first prototype built a local per-call order vector. Exact stage `1647664` preserved SHA-256 `3b87bb36da711bcef3ca877f3922e9fa5f8a3c6ae9d638c1ae18d4965bd92c5f` and halved helper cycles, but full Rank-27 pair `1647680` regressed 453.199 to 461.583 seconds and raised memory. That arm is rejected.

The accepted general repair adds one 32-bit rank per current Kernel variable. The external-rank vector remains permanent and its import-order vector remains the inverse. The new direct internal-rank image is initialized to `UINT32_MAX`, written with the same rank at first import, and compacted alongside every other per-current- variable vector. Hot current-variable comparisons read it directly. Only an unrecorded direct Kernel client takes the unchanged external-order fallback. The invariant

checker compares both views after every remap, and the Kernel regression pins the direct image before and after a compaction with holes. There is no change to candidate order, proof order, ticks, scheduling, learning, pruning, reconstruction identity, N1–N4, or the proof-rule/strategy boundary.

2 Exact state and long-horizon performance

Exact common-state job `1647683` preserves all 13 canonical rounds, all 250 rewrite rows, 335,924 images, and every tick and effect. Both arms have complete normalized non-time SHA-256 `3b87bb36da711bcef3ca877f3922e9fa5f8a3c6ae9d638c1ae18d4965bd92c5f`. The helper falls 448.814M to 302.824M cycles (−32.53%), snapshot time falls 199 to 186 ms, Substitute falls 1623 to 1599 ms, and complete prefix cycles fall 63.189B to 62.207B (−1.55%). Trace `1647685` emits the same 64 rows at SHA-256 `7a6216bbcdff016c361610991131a5407a8b4a2e8024480579bf4709dc237cc1`.

The predeclared six-instance checked screen `1647686/1647693` passes 12/12 with three SAT models and three UNSAT proofs per arm, exact paired non-time state, and zero residue. Its 1.01151 timing ratio is environmentally inconclusive: unrelated rule times swing in opposite directions on DivS and T105, while the other four cases add only 73 ms of Substitute time.

Dependency-serialized same-allocation Rank-27 job `1647694` gives the causal long control. Both arms return UNSAT exit 20 with accepted complete non-time SHA-256 `14f7da2d94153a2afd1776ecab8e7e49e5c2c0613dc26c43714471b7ecbe6194`. Internal/wall time improves 458.387/458.89 to 449.707/450.21 seconds. Of the 8.680-second gain, inprocessing contributes 0.940 seconds and work outside the inprocessing clock contributes 7.740; Substitute itself falls 23.576 to 23.504 seconds. This is a general cache effect, so downstream search/scaffolding can benefit even when the named Substitute timer moves only slightly.

User-requested same-allocation control `1647782` resolves the otherwise counterintuitive independent observation that a proof-producing run took 380.749 seconds while an earlier no-proof run took 450.21. On one node and allocation, the exact binary runs proof-on first in 405.375 seconds internal / 405.86 wall, then proof-off in 376.843/377.31 seconds. Proof generation therefore costs 28.532/28.55 seconds (+7.57%) even though proof-off receives the warm-state advantage. Checking afterward adds 106.37 seconds, making the complete proof pipeline 512.23 seconds. Proof-off and proof-on reproduce their respective accepted non-time hashes; all common logical state is exact, with only the expected proof-sink-dependent 432 versus 0 Sweep general-pair emission count. The proof reproduces its accepted hash and verifies. SLURM records the job as failed only because a final anchored text check did not accept `drat-trim`'s carriage-return-formatted `s VERIFIED` line after its zero exit; the cleanup trap still leaves zero residue. The prior timing inversion was cross-run environmental variation, not a proof-mode speedup.

Rank-27 peak RSS rises from 637,064 to 688,080 KiB, larger than the direct 1.7-MiB four-byte image. The broader six-instance screen rises at most 14.6 MiB, consistent with four bytes per current variable plus allocator effects; both remain far below 16 GiB. The Rank-27 RSS delta is retained as an anomaly, not silently attributed to the designed vector.

3 Complete correctness gates

Frozen full-width B28+C5 campaign `1647702/1647768` runs every task with two CPUs, 16 GiB, and the pre-Phase-4.5 1200-second solver contract. All 66 results are valid. Each arm independently validates 17 SAT models and 16 UNSAT proofs against the original decompressed CNF. Candidate/shipped PAR-2 is 1550.813/1537.768 seconds, ratio 1.008483, inside the predeclared 1.10 guard; peak RSS is 2.9317/2.9166 GiB. There is no missing or malformed task, unexplained solve loss, invalid model or proof, cleanup failure, or certificate residue.

Proof-producing immutable Rank-27 job `1647775` returns valid UNSAT in 380.749 seconds wall with 669,880 KiB peak RSS. Pinned `drat-trim` verifies all 1,514,395,111 proof bytes in 112.291 seconds with `s VERIFIED`. Proof SHA-256 is `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d`; normalized proof-mode non-time SHA-256 is `8cd3eda481799871a179e1483a57413622bc7832c09432df705b2308bafc2da7`. Both are byte-identical to the accepted active-decision witness. The proof, decompressed input, checker output, and scratch are deleted before atomic result publication, leaving zero certificate residue.

Claude Code exclusively prepares, submits, monitors, inspects, and classifies the exact-source ASan/UBSan native-LRAT campaign. Its authoritative terminal report is PASS. Prep job `1647778`, main leg `1647779`, and dedicated compaction-reach phase `1647781` preserve source-diff SHA-256 `681d17c986180b99b3c4cc6eb830dc32e88ea6a5b1ab2f1233872bca786ee1b6`. Corpus replay is 27/27 and random fuzz is 10,000/10,000 with zero failures. All 5,668 SAT models validate independently against their original CNFs; all 3,906 UNSAT native-LRAT proofs pass pinned `lrat-check`; pinned Kissat agrees on all 9,574 verdict cases. The 426 generated cases that hit the declared five-second quick bound are bounded semantic nonterminations with no oracle claim. A dedicated reach phase adds 8 checked SAT and 10 checked UNSAT cases, reaching VariableCompaction in 17/17 passing cases and Substitute in all 18. Claude reports no sanitizer finding, crash, invariant abort, model/proof rejection, differential disagreement, repair trigger, certificate residue, or job scratch residue. Earlier prep attempts `1647776`/`1647777` and wrapper-timeout attempt `1647780` are classified as non-solver events in Claude's frozen manifest; none carries a fuzz failure claim. Codex accepts the terminal report without inspecting fuzz artifacts.

4 Disposition and next work

Claude's PASS closes the final correctness gate. The evidence supports the maintained direct import-rank image as a portable Kernel representation repair rather than a benchmark arm. The 300-second Rank-27 target remains a milestone, not a stopping rule.

The next ordered increment is collection and rewatching. Freeze this accepted baseline and current-Kissat comparator, then trace watch compaction, reference forwarding, metadata reconstruction, sparse collection, and rewatching with work normalized by entries and literals. The first bounded hypothesis is a fused compact-and-forward traversal, gated first by exact common-prefix order, ticks, scheduling, learning, and pruning, then full Rank 27 and mechanism- relevant B+C, then the complete B28+C5 gate, independent proof/model checks, Claude-owned fuzz, and a separate report/commit. Substitute's remaining rewrite/collection cost follows that work; residual Congruence and broad search quality remain queued behind it.